# Lecture 1: Linear Algebra for Machine Learning

Pr. Ismail Merabet

Univ. of K-M-Ouargla

November 5, 2026

# Outline

1. Scalar, vector, matrix and tensor

# Outline

1. Scalar, vector, matrix and tensor
2. System of equations

# Outline


1. Scalar, vector, matrix and tensor
2. System of equations
3. Norms

# Outline


1. Scalar, vector, matrix and tensor
2. System of equations
3. Norms
4. Special Matrices and Vectors

# Outline


1. Scalar, vector, matrix and tensor
2. System of equations
3. Norms
4. Special Matrices and Vectors
5. Eigenvectors and eigenvalues

# Outline


1. Scalar, vector, matrix and tensor
2. System of equations
3. Norms
4. Special Matrices and Vectors
5. Eigenvectors and eigenvalues
6. Singular Value Decomposition

# Outline


1. Scalar, vector, matrix and tensor
2. System of equations
3. Norms
4. Special Matrices and Vectors
5. Eigenvectors and eigenvalues
6. Singular Value Decomposition
7. Moore-Penrose Pseudoinverse

# Scalars

- A scalar is a single number
- Integers, real numbers, rational numbers, etc.
- We denote it with italic font:

$$a, \quad n, \quad x$$

# Vectors

- A vector is a 1-D array of numbers:

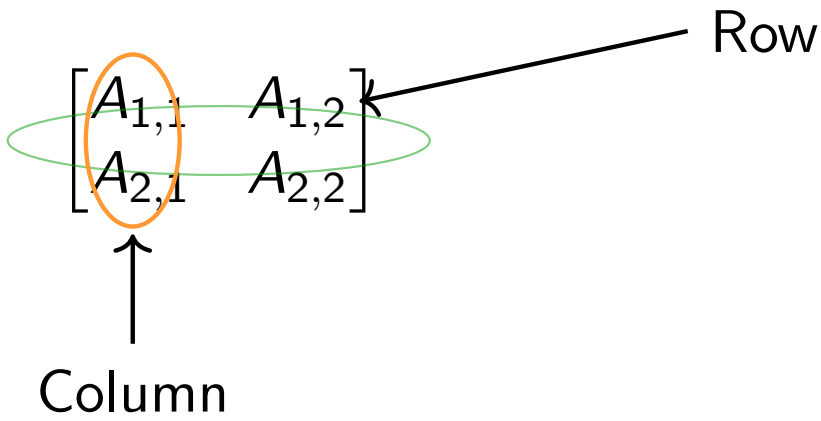
$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Can be real, binary, integer, etc.
- Example notation for type and size:

$$\boldsymbol{x} \in \mathbb{R}^n$$

# Matrices

- A matrix is a 2-D array of numbers:




- Example notation for type and shape:

$$\boldsymbol{A} \in \mathbb{R}^{m \times n}$$

# Matrices

- A matrix is a 2-D array of numbers:
- Example notation for type and shape:

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^\top = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

$$(\boldsymbol{A}^\top)_{i,j} = A_{j,i}$$

# Tensors

- A tensor is an array of numbers, that may have
  - zero dimensions, and be a scalar
  - one dimension, and be a vector
  - two dimensions, and be a matrix
  - or more dimensions.

# Matrix Transpose

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^\top = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

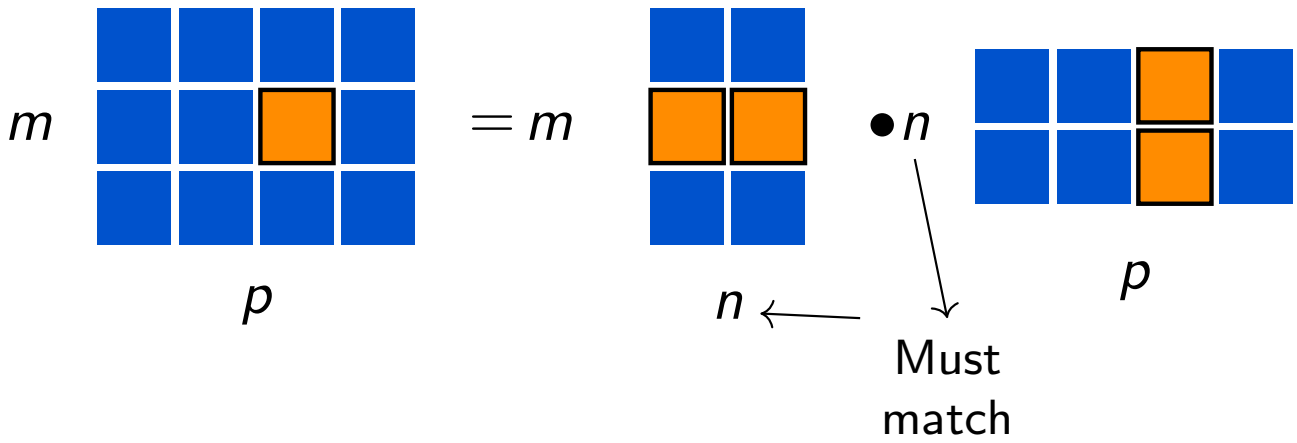
Figure 2.1: The transpose of the matrix can be thought of as a mirror image across the main diagonal.

$$(\boldsymbol{A}^\top)_{i,j} = A_{j,i} \qquad (\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$$

# Matrix (Dot) Product

$$\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}. \tag{1}$$

$$C_{i,j} = \sum_k A_{i,k} B_{k,j}. \tag{2}$$



# Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Figure 2.2: Example identity matrix: This is $\boldsymbol{I}_3$.

$$\forall \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{I}_n \boldsymbol{x} = \boldsymbol{x}.$$

# Systems of Equations

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \tag{3}$$

expands to

$$\boldsymbol{A}_{1,:}\boldsymbol{x} = b_1 \tag{4}$$
$$\boldsymbol{A}_{2,:}\boldsymbol{x} = b_2 \tag{5}$$
$$\vdots$$
$$\boldsymbol{A}_{m,:}\boldsymbol{x} = b_m \tag{6}$$

# Systems of Equations

$$A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n = b_1 \tag{7}$$
$$A_{2,1}x_1 + A_{2,2}x_2 + \cdots + A_{2,n}x_n = b_2 \tag{8}$$
$$\vdots$$
$$A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n = b_m \tag{9}$$

# Solving Systems of Equations

- A linear system of equations can have:
  - No solution
  - Many solutions
  - Exactly one solution: this means multiplication by the matrix is an invertible function

# Matrix Inversion

- Matrix inverse:

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_n$$

- Solving a system using an inverse:

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b} \\ \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \\ \boldsymbol{I}_n\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \end{aligned}$$

- Numerically unstable, but useful for abstract analysis

# Invertibility

- Matrix can't be inverted if...
  - More rows than columns
  - More columns than rows
  - Redundant rows/columns ("linearly dependent", "low rank")

# Norms

- Functions that measure how "large" a vector is
- Similar to a distance between zero and the point represented by the vector

$$\|\boldsymbol{x}\|_p = \left( \sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- A norm is any function $f$ satisfying:
  - $f(\boldsymbol{x}) = 0 \Rightarrow \boldsymbol{x} = \boldsymbol{0}$
  - $f(\boldsymbol{x} + \boldsymbol{y}) \leq f(\boldsymbol{x}) + f(\boldsymbol{y})$ (the triangle inequality)
  - $\forall \alpha \in \mathbb{R},\ f(\alpha \boldsymbol{x}) = |\alpha| f(\boldsymbol{x})$

# Norms

- $L^p$ norm

$$\|\boldsymbol{x}\|_p = \left( \sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- Most popular norm: $L^2$ norm, $p = 2$
- $L^1$ norm, $p = 1$:

$$\|\boldsymbol{x}\|_1 = \sum_i |x_i|.$$

- Max norm, infinite $p$:

$$\|\boldsymbol{x}\|_\infty = \max_i |x_i|.$$

# Norms (for matrices)

1. Induced norms: Defined by

$$\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}.$$

   - $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (largest absolute column sum).
   - $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (largest absolute row sum).
   - $\|A\|_2 = \sigma_1 = \sqrt{\lambda_{\max}(A^* A)}$ (spectral norm).

2. Frobenius norm:

$$\|\boldsymbol{A}\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$$

# Special Matrices and Vectors

- Unit vector:

$$\|\boldsymbol{x}\|_2 = 1.$$

- Symmetric Matrix:

$$\boldsymbol{A} = \boldsymbol{A}^\top.$$

- Orthogonal matrix:

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I}$$
$$\boldsymbol{A}^{-1} = \boldsymbol{A}^\top$$

# Eigendecomposition

- Eigenvector and eigenvalue:

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}.$$

- Eigendecomposition of a diagonalizable matrix:

$$\boldsymbol{A} = \boldsymbol{V}\,\mathrm{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}.$$

- Every real symmetric matrix has a real, orthogonal eigendecomposition:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{\top}$$

# Eigendecomposition Example 1: Eigenvalues

Consider the symmetric matrix

$$\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

# Eigendecomposition Example 1: Eigenvalues

Consider the symmetric matrix

$$\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

**Step 1: Find the eigenvalues.**

$$\begin{aligned} \det(\boldsymbol{A} - \lambda \boldsymbol{I}) &= (2-\lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 \\ &= (\lambda - 3)(\lambda - 1) = 0. \end{aligned}$$

Hence $\lambda_1 = 3$ and $\lambda_2 = 1$.

# Eigendecomposition Example 1: Eigenvectors

**Step 2: Solve $(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{q} = 0$.**

$$\lambda_1 = 3: \ y = x, \qquad \lambda_2 = 1: \ y = -x.$$

Normalize the eigenvectors:

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

# Eigendecomposition Example 1: Eigenvectors

**Step 2: Solve $(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{q} = 0$.**

$$\lambda_1 = 3: \ y = x, \qquad \lambda_2 = 1: \ y = -x.$$

Normalize the eigenvectors:

$$\boldsymbol{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

**Step 3: Assemble and verify the decomposition.**

$$\boldsymbol{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \boldsymbol{\Lambda} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$\boxed{\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top = \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}}$$

# Eigendecomposition Example 2: A Nonsymmetric Matrix

$$\boldsymbol{B} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \qquad \det(\boldsymbol{B} - \lambda \boldsymbol{I}) = (2 - \lambda)(3 - \lambda).$$

Solve for eigenvectors:

# Eigendecomposition Example 2: A Nonsymmetric Matrix

$$\boldsymbol{B} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, \qquad \det(\boldsymbol{B} - \lambda \boldsymbol{I}) = (2 - \lambda)(3 - \lambda).$$

Solve for eigenvectors:

$$\lambda_1 = 2: \ \boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \lambda_2 = 3: \ \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$\boldsymbol{V} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \boldsymbol{V}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

$$\boxed{\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \boldsymbol{B}}$$

The eigenvectors are independent, but not orthogonal: $\boldsymbol{V}^{-1} \neq \boldsymbol{V}^{\top}$.

# Singular Value Decomposition

- Similar to eigendecomposition
- More general; matrix need not be square

$$\text{Eigendecomposition:} \quad \boldsymbol{A} = \boldsymbol{V}\,\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}$$

$$\text{SVD:} \quad \boldsymbol{A} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\top}$$

If $\boldsymbol{A}$ is $m \times n$, then $\boldsymbol{U}$ is $m \times m$, $\boldsymbol{D}$ is $m \times n$, and $\boldsymbol{V}$ is $n \times n$.

# SVD Example 1: Singular Values

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{A}^\top \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

# SVD Example 1: Singular Values

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{A}^\top \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

**Step 1: Find the eigenvalues of $\boldsymbol{A}^\top \boldsymbol{A}$.**

$$\det(\boldsymbol{A}^\top \boldsymbol{A} - \lambda \boldsymbol{I}) = \lambda^2 - 3\lambda + 1 = 0.$$

$$\lambda_1 = \frac{3 + \sqrt{5}}{2}, \qquad \lambda_2 = \frac{3 - \sqrt{5}}{2}.$$

# SVD Example 1: Singular Values

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{A}^\top \boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

**Step 1: Find the eigenvalues of $\boldsymbol{A}^\top \boldsymbol{A}$.**

$$\det(\boldsymbol{A}^\top \boldsymbol{A} - \lambda \boldsymbol{I}) = \lambda^2 - 3\lambda + 1 = 0.$$

$$\lambda_1 = \frac{3 + \sqrt{5}}{2}, \qquad \lambda_2 = \frac{3 - \sqrt{5}}{2}.$$

**Step 2: Take the nonnegative square roots.**

$$\sigma_1 = \frac{1 + \sqrt{5}}{2} \approx 1.618, \qquad \sigma_2 = \frac{\sqrt{5} - 1}{2} \approx 0.618.$$

$$\boldsymbol{D} = \operatorname{diag}(\sigma_1, \sigma_2).$$

# SVD Example 1: Singular Vectors

**Step 3: Normalize eigenvectors of $\boldsymbol{A}^\top \boldsymbol{A}$.** Define $\varphi = (1+\sqrt{5})/2$ and $N = \sqrt{1+\varphi^2}$.

$$\boldsymbol{v}_1 = \frac{1}{N}\begin{bmatrix}1\\ \varphi\end{bmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{N}\begin{bmatrix}\varphi\\ -1\end{bmatrix}.$$

# SVD Example 1: Singular Vectors

**Step 3: Normalize eigenvectors of $\boldsymbol{A}^\top \boldsymbol{A}$.** Define $\varphi = (1+\sqrt{5})/2$ and $N = \sqrt{1+\varphi^2}$.

$$\boldsymbol{v}_1 = \frac{1}{N}\begin{bmatrix}1\\ \varphi\end{bmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{N}\begin{bmatrix}\varphi\\ -1\end{bmatrix}.$$

**Step 4: Compute $\boldsymbol{u}_i = \boldsymbol{A}\boldsymbol{v}_i/\sigma_i$.**

$$\boldsymbol{u}_1 = \frac{1}{N}\begin{bmatrix}\varphi\\ 1\end{bmatrix}, \qquad \boldsymbol{u}_2 = \frac{1}{N}\begin{bmatrix}1\\ -\varphi\end{bmatrix}.$$

Thus

$$\boldsymbol{U} = \frac{1}{N}\begin{bmatrix}\varphi & 1\\ 1 & -\varphi\end{bmatrix}, \qquad \boldsymbol{V} = \frac{1}{N}\begin{bmatrix}1 & \varphi\\ \varphi & -1\end{bmatrix}.$$

Both matrices have orthonormal columns.

# SVD Example 1: Reconstruction

**Step 5: Assemble $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top$.**

$$\boldsymbol{U} \approx \begin{bmatrix} 0.851 & 0.526 \\ 0.526 & -0.851 \end{bmatrix}, \qquad \boldsymbol{D} \approx \begin{bmatrix} 1.618 & 0 \\ 0 & 0.618 \end{bmatrix}.$$

$$\boldsymbol{V}^\top \approx \begin{bmatrix} 0.526 & 0.851 \\ 0.851 & -0.526 \end{bmatrix}.$$

Using the exact factors from the previous slide,

$$\boxed{\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \boldsymbol{A}}.$$

- The displayed decimals reconstruct $\boldsymbol{A}$ up to rounding.
- The two positive singular values show that $\operatorname{rank}(\boldsymbol{A}) = 2$.

# SVD Example 2: A Rectangular Matrix

$$\boldsymbol{B} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{B}^\top \boldsymbol{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Eigenvalues $2, 0$ give singular values $\sqrt{2}, 0$ and

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Since $\boldsymbol{u}_1 = \boldsymbol{B}\boldsymbol{v}_1/\sqrt{2} = (1, 0, 0)^\top$, complete an orthonormal basis with the other coordinate vectors:

$$\boldsymbol{U} = \boldsymbol{I}_3, \quad \boldsymbol{D} = \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \boldsymbol{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$\boxed{\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^\top = \boldsymbol{B}} \qquad \operatorname{rank}(\boldsymbol{B}) = 1.$$

# Moore-Penrose Pseudoinverse

$A^+$ is the unique matrix satisfying

$$AA^+A = A, \tag{10}$$
$$A^+AA^+ = A^+, \tag{11}$$
$$(AA^+)^T = AA^+, \tag{12}$$
$$(A^+A)^T = A^+A. \tag{13}$$

- If the equation

$$\boldsymbol{x} = \boldsymbol{A}^+\boldsymbol{y}$$

has:

  - Exactly one solution: this is the same as the inverse.
  - No solution: this gives us the solution with the smallest error $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_2$.
  - Many solutions: this gives us the solution with the smallest norm of $\boldsymbol{x}$.

# Useful properties

- $(A^+)^+ = A$.
- $(A^T)^+ = (A^+)^T$.
- If $A$ is square and invertible, then $A^+ = A^{-1}$.
- For a scalar or a diagonal matrix, invert the nonzero entries and leave the zeros alone.
- **Caution:** $(AB)^+ \neq B^+A^+$ in general.

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

(rank 1). Then

$$A^+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The ordinary inverse does not exist, but the pseudoinverse inverts the part that can be inverted and ignores the rest.
If we want to solve $Ax = b$, with $b = (1, 1)^T$, then

$$x^* = A^+ b = \arg\min \|Ax - b\|_2$$

# Example

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

It has full row rank , so we can use $A^+ = A^T(AA^T)^{-1}$.

$$AA^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

$$A^+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{pmatrix}.$$

- $AA^+ = I_2$, since $A$ has full row rank, so $A^+$ is a right inverse.

# Example

Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $AB = I_2$, we have

$$(AB)^+ = I_2.$$

The Moore–Penrose pseudoinverses are

$$A^+ = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}, \qquad B^+ = (B^{\mathsf{T}}B)^{-1}B^{\mathsf{T}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{(AB)^+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = B^+A^+.}$$

# Computing the Pseudoinverse

The SVD allows the computation of the pseudoinverse:

$$\boldsymbol{A}^{+} = \boldsymbol{V}\boldsymbol{D}^{+}\boldsymbol{U}^{\top}$$

Take reciprocal of non-zero entries

Equivalently, the pseudoinverse is defined as

$$\boldsymbol{A}^{+} = \lim_{\alpha \searrow 0} (\boldsymbol{A}^{\top}\boldsymbol{A} + \alpha \boldsymbol{I})^{-1}\boldsymbol{A}^{\top}$$

where $\boldsymbol{U}, \boldsymbol{D}, \boldsymbol{V}$ are the SVD of $\boldsymbol{A}$ and $\boldsymbol{D}^{+}$ is obtained by taking the reciprocal of the non-zero elements of $\boldsymbol{D}$ and then transposing the result.

# Trace

$$\mathrm{Tr}(\boldsymbol{A}) = \sum_i A_{i,i}.$$

$$\mathrm{Tr}(\boldsymbol{A}) = \mathrm{Tr}(\boldsymbol{A}^\top)$$

$$\mathrm{Tr}(\boldsymbol{ABC}) = \mathrm{Tr}(\boldsymbol{CAB}) = \mathrm{Tr}(\boldsymbol{BCA})$$

# Learning linear algebra

- Do a lot of practice problems
- Start out with lots of summation signs and indexing into individual entries
- Eventually you will be able to mostly use matrix and vector product notation quickly and easily